DEPARTMENT OF MECHATRONICS ENGINEERING

MINI PROJECT REPORT

AUTOMATED TANK WATER LEVEL CONTROL SYSTEM

EMD51002 EMBEDDED SYSTEM FOR MECHATRONICS

SUBMITTED BY:

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INTRODUCTION

Arduino-based automatic water level control system project we are going to measure the water level by using ultrasonic sensors. The basic principle of ultrasonic distance measurement is based on ECHO. When sound waves are transmitted in the environment they return to the origin as ECHO after striking on any obstacle. So we have to only calculate the traveling time of both sounds means outgoing time and returning time to origin after striking any obstacle and after some calculation, we can get a result that is the distance. This concept is used in our water controller project where the water motor pump is automatically turned on when the water level in the tank becomes low. Then the value of input is displayed on the LCD .

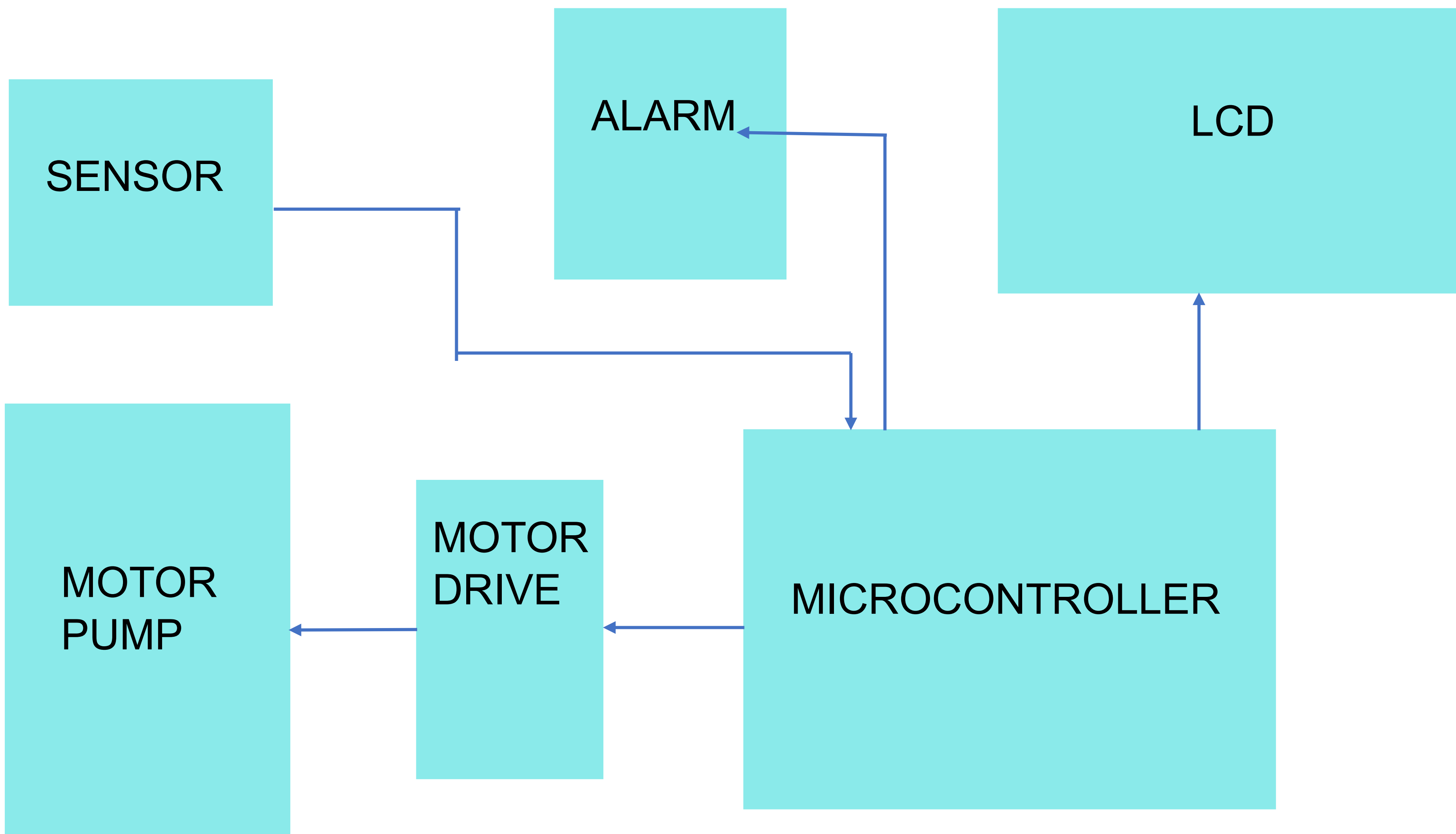
Required components :

- Arduino Uno
- Ultrasonic sensor Module
- 16x2 LCD
- Relay 6 Volt
- Breadboard
- Hoop up wires
- Jumper wire
- Buzzer
- Motor pump

OBJECTIVES

The Arduino-based water level control system automates water pump operation based on tank levels, conserving water, preventing pump damage, and offering smart irrigation. It provides a user-friendly interface with an LCD, customizable thresholds, and promoting water conservation and efficient usage in agricultural and domestic settings.

BLOCK DIAGRAM AND EXPLANATION:



MICROCONTROLLER

The Arduino Uno R3 is a microcontroller board based on a removable, dual-inline-package (DIP) ATmega328 AVR microcontroller. It has 20 digital input/output pins (of which 6 can be used as PWM outputs and 6 can be used as analog inputs). Programs can be loaded onto it from the easy-to-use Arduino computer program. The Arduino has an extensive support community, which makes it a very easy way to get started working with embedded electronics. The R3 is the third, and latest, revision of the Arduino Uno.

PINS

Digital Pins: There are 14 digital pins on Arduino Uno which can be connected to components like LED, LCD, etc.

Analog Pins: There are 6 analog pins on the Uno. These pins are generally used to connect sensors because all the sensors generally have analog values. Most of the input components are connected here.

Power Supply: The power supply pins are IOREF, GND, 3.3V, 5V, and Vin are used to connect sensors because all the sensors generally have analog values. Most of the input components are connected here.

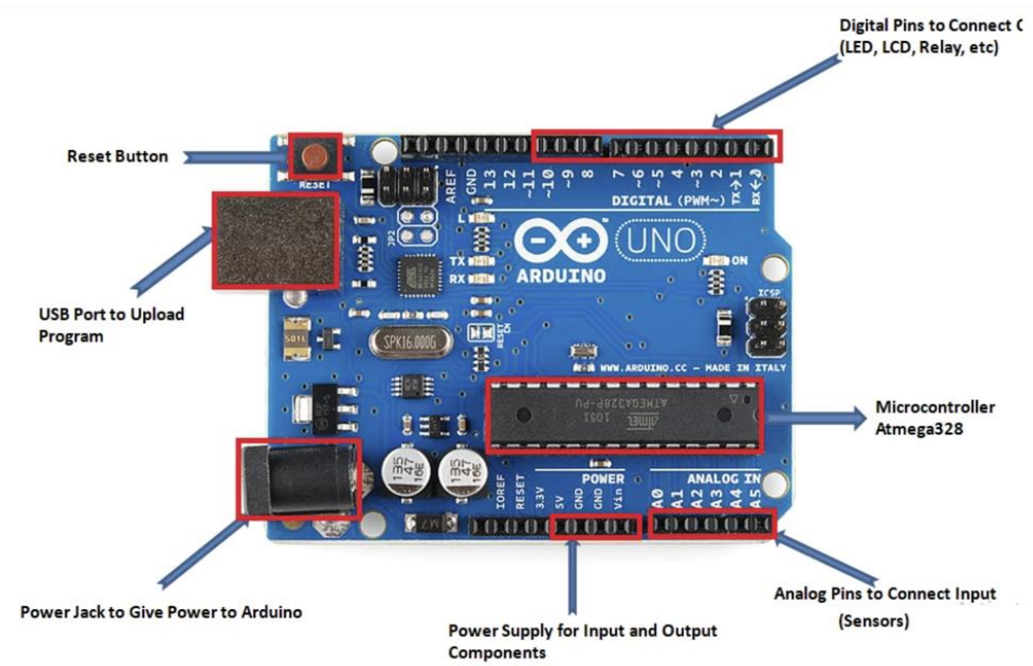
Power Jack: Uno board can be powered both by external supply and via USB cable.

USB Port: This port function is to program the board or to upload the program. The program can be uploaded to the board with the help of Arduino IDE and USB cable.

Reset Button: This is used to restart the uploaded program.

General specifications

Processor:	ATmega328 @ 16 MHz
RAM size:	2048 bytes
Program memory size:	31.5 Kbytes
Motor channels:	0
User I/O lines:	20[1]
Max current on a single I/O:	40 mA
Minimum operating voltage:	7 V
Maximum operating voltage:	12 V

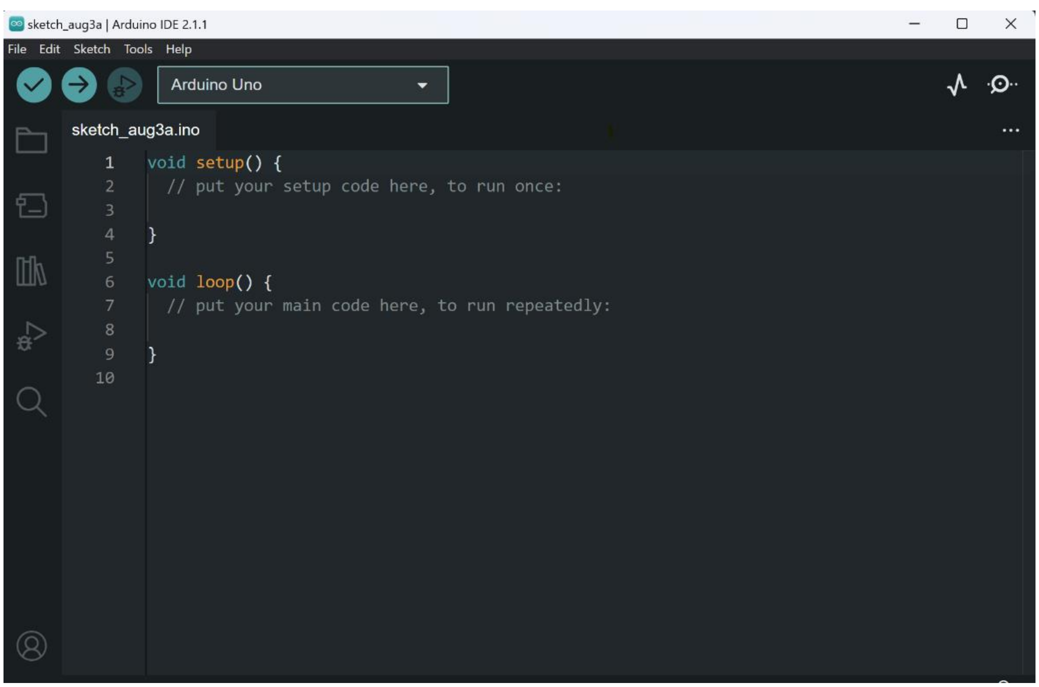


Arduino IDE

To program the Arduino boards, users utilize the Arduino Integrated Development Environment (IDE), which is a user-friendly software application. The Arduino IDE provides an easy-to-use interface for writing, compiling, and uploading code to the Arduino board. It simplifies the programming process, making it accessible to beginners and experienced developers alike.

The Arduino programming language is based on a simplified version of C/C++ and is easy to understand and learn. Users write their code in the Arduino IDE, and the software translates it into machine code that the microcontroller can execute.

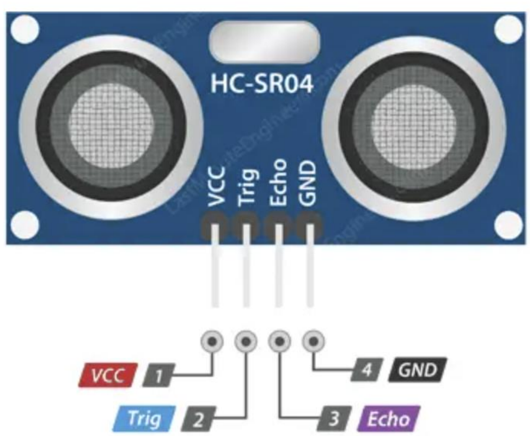
The microcontrollers can be programmed using the C and C++ programming languages(Embedded C), using a standard API which is also known as the Arduino Programming Language.



ULTRASONIC SENSOR -HC-SR04

An HC-SR04 ultrasonic distance sensor consists of two ultrasonic transducers. One acts as a transmitter that converts the electrical signal into 40 KHz ultrasonic sound pulses. The other acts as a receiver and listens for the transmitted pulses. When the receiver receives these pulses, it produces an output pulse whose width is proportional to the distance of the object in front. This sensor provides excellent non-contact range detection between 2 cm to 400 cm (~13 feet) with an accuracy of 3 mm. Since it operates on 5 volts, it can be connected directly to an Arduino or any other 5V logic microcontroller.

Working of HC-SR04



VCC supplies power to the HC-SR04 ultrasonic sensor. You can connect it to the 5V output from your Arduino.

A trig (Trigger) pin is used to trigger ultrasonic sound pulses. By setting this pin to HIGH for 10µs, the sensor initiates an ultrasonic burst.

The echo pin goes high when the ultrasonic burst is transmitted and remains high until the sensor receives an echo, after which it goes low. By measuring the time the Echo pin stays high, the distance can be calculated.

GND is the ground pin. Connect it to the ground of the Arduino.

It all starts when the trigger pin is set HIGH for 10µs. In response, the sensor transmits an ultrasonic burst of eight pulses at 40 kHz. This 8-pulse pattern is specially designed so that the receiver can distinguish the transmitted pulses from ambient ultrasonic noise.

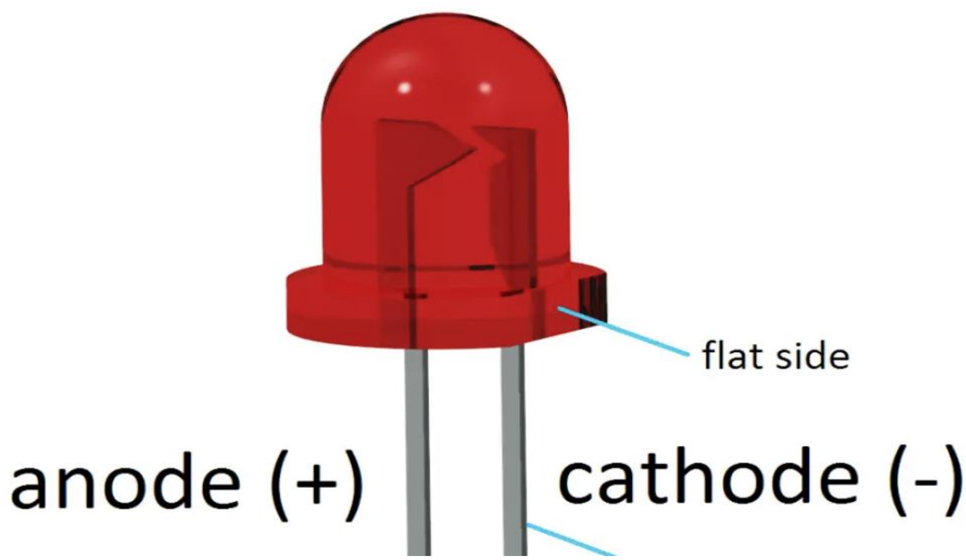
These eight ultrasonic pulses travel through the air away from the transmitter. Meanwhile, the echo pin goes HIGH to initiate the echo-back signal.

If those pulses are not reflected, the echo signal times out and goes low after 38ms (38 milliseconds). Thus a pulse of 38ms indicates no obstruction within the range of the sensor.

If those pulses are reflected, the echo pin goes low as soon as the signal is received. This generates a pulse on the echo pin whose width varies from 150 µs to 25 ms depending on the time taken to receive the signa

LED

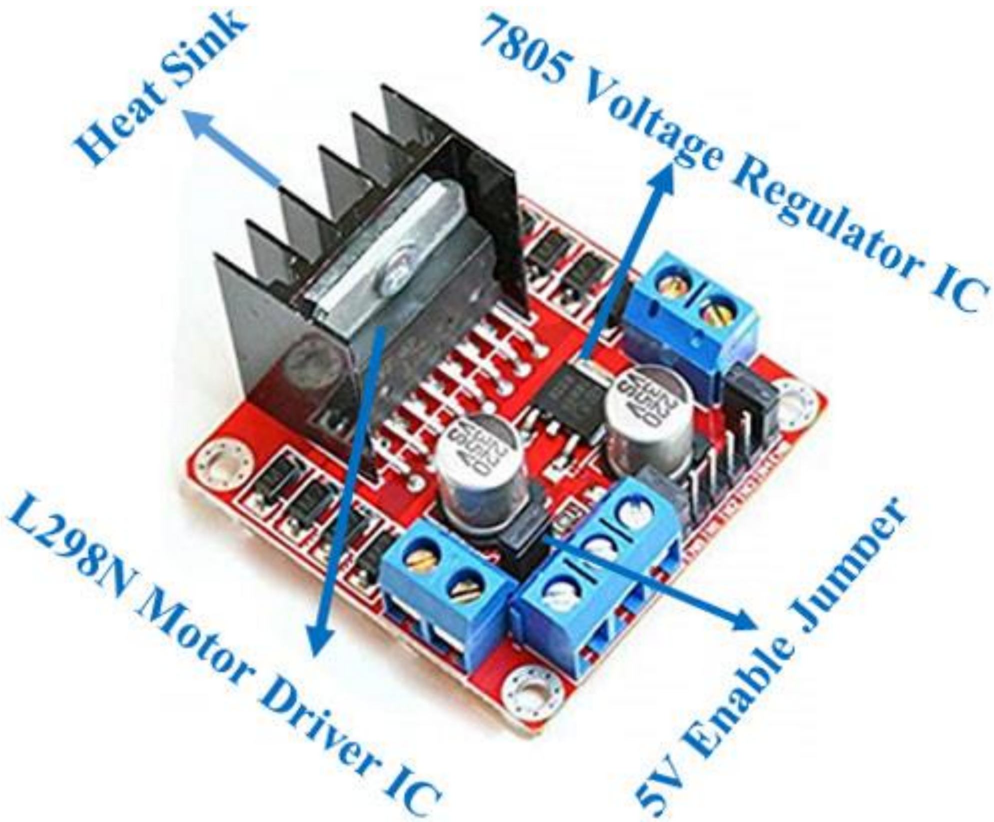
A **light-emitting diode** (**LED**) is a semiconductor device that emits light when current flows through it. Electrons in the semiconductor recombine with electron holes, releasing energy in the form of photons. The color of the light (corresponding to the energy of the photons) is determined by the energy required for electrons to cross the band gap of the semiconductor. White light is obtained by using multiple semiconductors or a layer of light-emitting phosphor on the semiconductor device



LEDs have many advantages over incandescent light sources, including lower power consumption, longer lifetime, improved physical robustness, smaller size, and faster switching. In exchange for these generally favorable attributes, disadvantages of LEDs include electrical limitations to low voltage and generally to DC (not AC) power, inability to provide steady illumination from a pulsing DC or an AC electrical supply source, and lesser maximum operating temperature and storage temperature

MOTOR DRIVER

The L298N Motor Driver module consists of an L298 Motor Driver IC, 78M05 Voltage Regulator, resistors, capacitor, Power LED, 5V jumper in an integrated circuit.



78M05 Voltage regulator will be enabled only when the jumper is placed. When the power supply is less than or equal to 12V, then the internal circuitry will be powered by the voltage regulator and the 5V pin can be used as an output pin to power the microcontroller. The jumper should not be placed when the power supply is greater than 12V and separate 5V should be given through 5V terminal to power the internal circuitry.

ENA & ENB pins are speed control pins for Motor A and Motor B while IN1& IN2 and IN3 & IN4 are direction control pins for Motor A and Motor B.

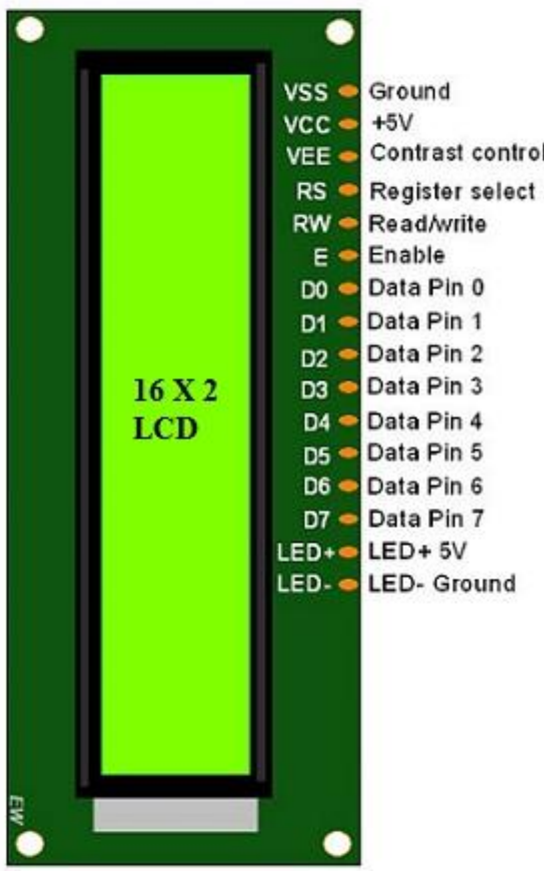
LCD

LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, has animations, and there are no limitations for displaying custom characters, special and even animations, etc.

The 16×2 LCD pinout is shown below.

- Pin1 (Ground/Source Pin): This is a GND pin of display, used to connect the GND terminal of the microcontroller unit or power source.
- Pin2 (VCC/Source Pin): This is the voltage supply pin of the display, used to connect the supply pin of the power source.
- Pin3 (V0/VEE/Control Pin): This pin regulates the difference of the display, used to connect a changeable POT that can supply 0 to 5V.
- Pin4 (Register Select/Control Pin): This pin toggles among command or data register, is used to connect a microcontroller unit pin, and obtains either 0 or 1(0 = data mode, and 1 = command mode).
- Pin5 (Read/Write/Control Pin): This pin toggles the display among the read or writes operation, and it is connected to a microcontroller unit pin to get either 0 or 1 (0 = Write Operation and 1 = Read Operation).

- Pin 6 (Enable/Control Pin): This pin should be held high to execute the Read/Write process, and it is connected to the microcontroller unit & constantly held high.
- Pins 7-14 (Data Pins): These pins are used to send data to the display. These pins are connected in two-wire modes like 4-wire mode and 8-wire mode. In the 4-wire mode, only four pins are connected to the microcontroller unit like 0 to 3, whereas in the 8-wire mode, 8-pins are connected to the microcontroller unit like 0 to 7.
- Pin15 (+ve pin of the LED): This pin is connected to +5V
- Pin 16 (-ve pin of the LED): This pin is connected to GND.



MOTOR PUMP

A 3V DC horizontal micro submersible pump is a small, low-cost pump that can be operated with a 2.5–6V power supply. It can move up to 120 liters of water per hour and has a low current consumption of 220mA.

To use the pump, connect a tube pipe to the motor outlet, submerge the pump in water, and power it. The pump works by sucking water through its inlet and releasing it through the outlet.

Here are some 3V DC horizontal micro submersible pumps:

DC 3-6 V Mini Micro Submersible Water Pump: A low-cost, small-size pump with a current consumption of 220mA

Micro Submersible Water Pump DC 3V-5V: Can be easily integrated into a water system project

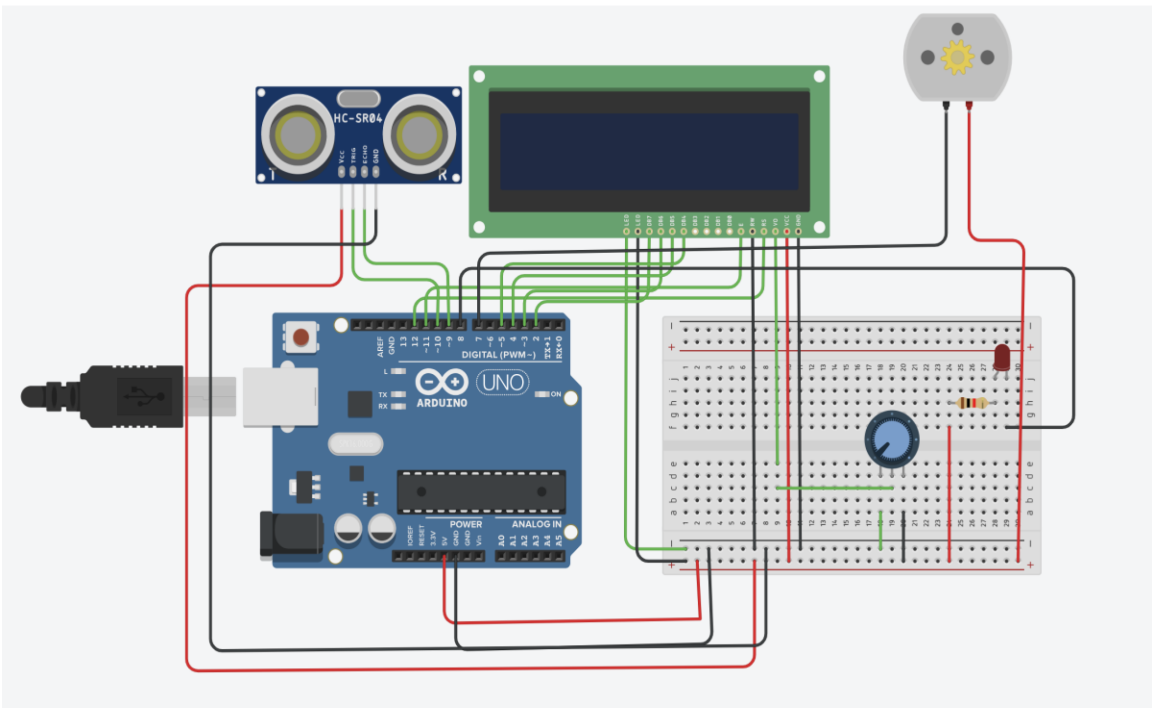
Submersible 3V DC Water Pump with 1 Meter Wire: A horizontal type pump that's great for beginner projects

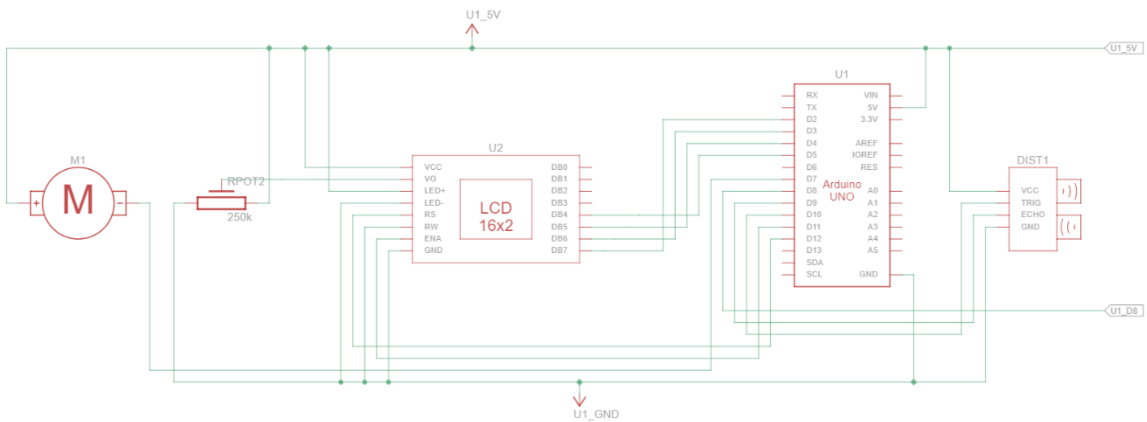
1pcs 3V - 12V DC Submersible Water Pump: A horizontal micropump with a flow rate of 150 liters per hour

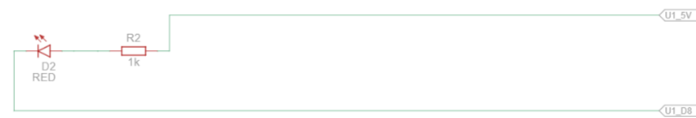
WORKING OF AUTOMATIC WATER LEVEL CONTROL SYSTEM

Arduino-based automatic water level control system project we are going to measure the water level by using ultrasonic sensors. The basic principle of ultrasonic distance measurement is based on ECHO. When sound waves are transmitted in the environment they return to their origin as ECHO after striking any obstacle.. This concept is used in our water controller project where the water motor pump is automatically turned on when the water level in the tank becomes low. Then the value of input is displayed on the LCD. Connections: The TRIG pin receives a trigger signal from the Arduino. The ECHO pin sends an echo signal back to the Arduino based on the reflected ultrasonic waves. It will display the current level of water in LCD. Relay receives a control signal from the Arduino to either open or close the circuit. It controls the power supply to the motor pump based on the Arduino's signal. Buzzer produces sound alerts based on the Arduino's signal (e.g. when the pump is turned on or off). Motor Pump receives power from the relay, controlled by the Arduino. The pump operates or stops based on the relay's status, which is controlled by the Arduino.

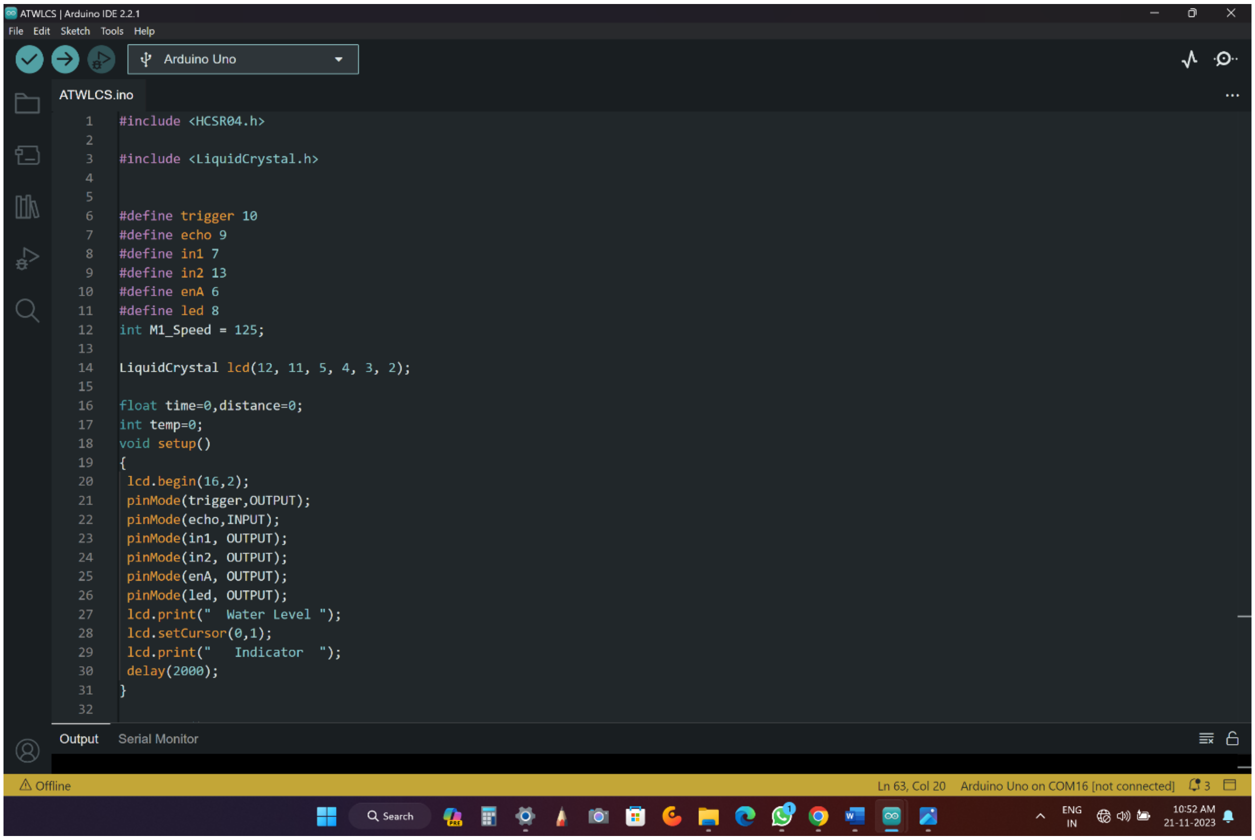
CIRCUIT DIAGRAM:



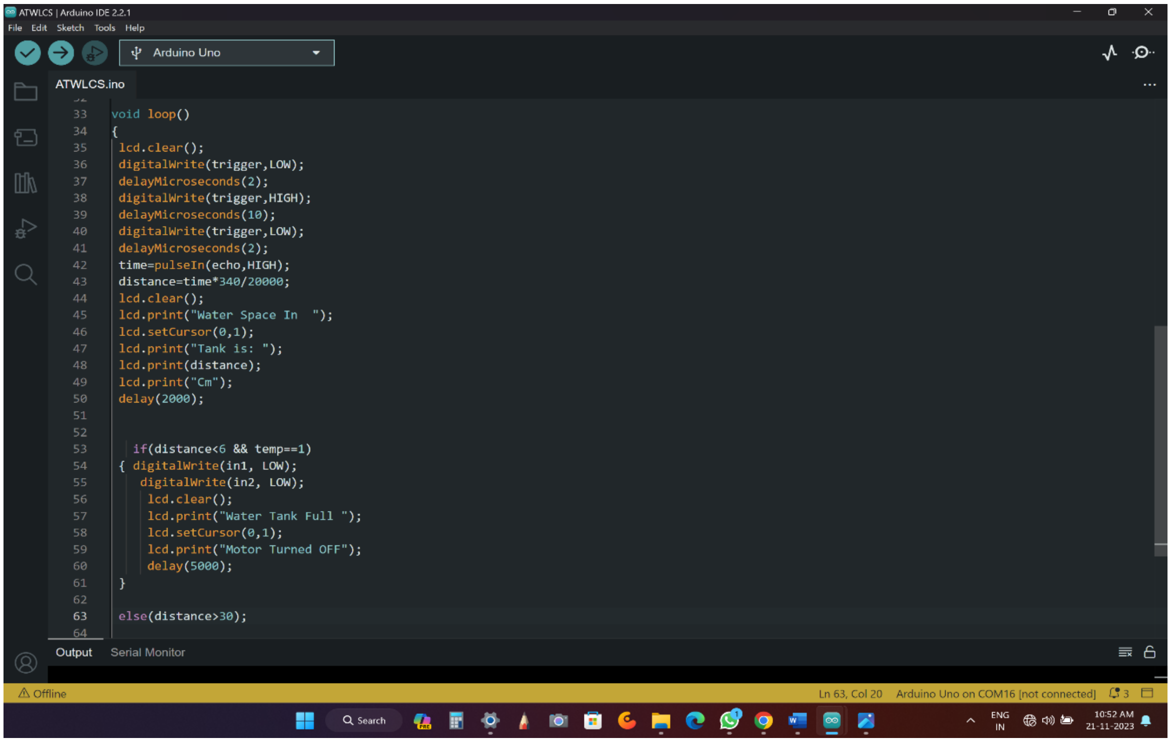

U1_5V
U1
Arduino UNO
U2
LCD 16x2
M1
M
RPOT2
250k
DIST1
U1_D8
U1_GND


U1_5V
R2
1k
D2
RED
U1_D8

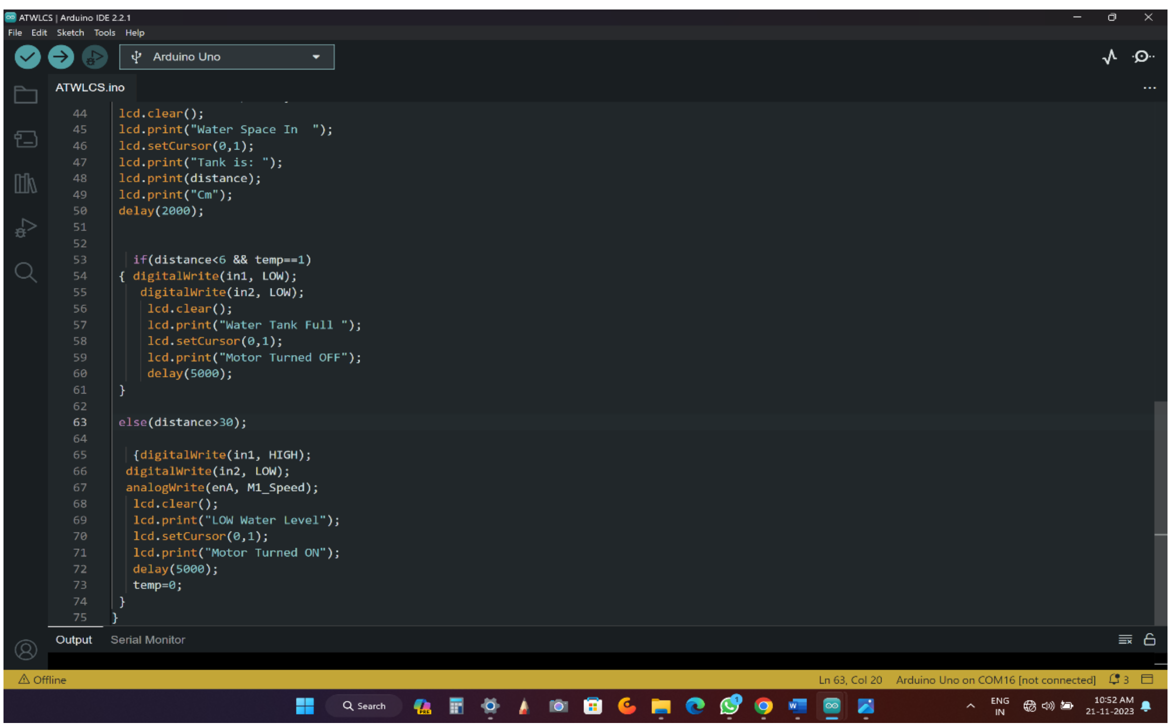
CODE



```
#include <HCSR04.h>

#include <LiquidCrystal.h>


#define trigger 10
#define echo 9
#define in1 7
#define in2 13
#define enA 6
#define led 8
int M1_Speed = 125;

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

float time=0,distance=0;
int temp=0;
void setup()
{
 lcd.begin(16,2);
 pinMode(trigger,OUTPUT);
 pinMode(echo,INPUT);
 pinMode(in1, OUTPUT);
 pinMode(in2, OUTPUT);
 pinMode(enA, OUTPUT);
 pinMode(led, OUTPUT);
 lcd.print("  Water Level ");
 lcd.setCursor(0,1);
 lcd.print("   Indicator  ");
 delay(2000);
}

```



```
void loop()
{
 lcd.clear();
 digitalWrite(trigger,LOW);
 delayMicroseconds(2);
 digitalWrite(trigger,HIGH);
 delayMicroseconds(10);
 digitalWrite(trigger,LOW);
 delayMicroseconds(2);
 time=pulseIn(echo,HIGH);
 distance=time*340/20000;
 lcd.clear();
 lcd.print("Water Space In  ");
 lcd.setCursor(0,1);
 lcd.print("Tank is: ");
 lcd.print(distance);
 lcd.print("Cm");
 delay(2000);


   if(distance<6 && temp==1)
 { digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
     lcd.clear();
     lcd.print("Water Tank Full ");
     lcd.setCursor(0,1);
     lcd.print("Motor Turned OFF");
     delay(5000);
 }

 else(distance>30);
```



```
 lcd.clear();
 lcd.print("Water Space In  ");
 lcd.setCursor(0,1);
 lcd.print("Tank is: ");
 lcd.print(distance);
 lcd.print("Cm");
 delay(2000);


   if(distance<6 && temp==1)
 { digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
     lcd.clear();
     lcd.print("Water Tank Full ");
     lcd.setCursor(0,1);
     lcd.print("Motor Turned OFF");
     delay(5000);
 }

 else(distance>30);

   {digitalWrite(in1, HIGH);
  digitalWrite(in2, LOW);
  analogWrite(enA, M1_Speed);
   lcd.clear();
   lcd.print("LOW Water Level");
   lcd.setCursor(0,1);
   lcd.print("Motor Turned ON");
   delay(5000);
   temp=0;
 }
}
```

RESULT

The Arduino-based water level control system utilizes ultrasonic sensors to measure water levels. When the water level is low, the Arduino triggers the motor pump through a relay, displayed on an LCD. The system integrates echo-based distance measurement, creating an automated solution for efficient water management and remote monitoring.

INDIVIDUAL CONTRIBUTIONS

1.	**THIRISHA.R**	**CIRCUIT DESIGN(CODE)**
2.	**SANDHIYA.R**	**CODE**
3.	**HARMITHA C.K**	**CIRCUIT DESIGN**